Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Optimization in Smart Cities

09.01.2026

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1. Introduction

Urban traffic congestion remains one of the most persistent challenges in modern cities, leading to increased travel time, fuel consumption, and environmental impact. Rapid urbanization, combined with rising vehicle ownership, has made traditional traffic management systems increasingly inadequate.

Cities such as those in the United Arab Emirates (UAE) highlight this paradox—despite advanced infrastructure and smart city initiatives, congestion continues to grow due to dynamic and unpredictable traffic patterns.

This research aims to explore how **Multi-Agent Reinforcement Learning (MARL)** can enable decentralized, adaptive traffic control systems that respond intelligently to real-time traffic conditions.

2. Problem Statement

Existing traffic signal systems are limited in their ability to adapt to real-world complexity.

Key challenges include:

- Static or semi-adaptive signal timing mechanisms
- Centralized decision-making architectures
- Limited coordination between intersections
- Inability to respond effectively to sudden traffic fluctuations

These limitations result in:

- Increased waiting time at intersections
- Inefficient traffic flow
- Higher fuel consumption and emissions
- Poor scalability in dense urban environments

There is a need for a **self-learning, decentralized system** capable of optimizing traffic flow dynamically.

3. Research Objectives

The primary objective of this research is to design and evaluate a **Multi-Agent Reinforcement Learning-based traffic signal control system**.

Specific Goals:

- Develop a simulation-based environment for urban traffic modeling
- Implement a baseline fixed-time traffic signal system
- Design decentralized agents for each intersection
- Train agents using reinforcement learning techniques
- Compare MARL-based control with traditional systems
- Evaluate scalability and adaptability under varying traffic conditions

4. Literature Gap

Existing research in traffic optimization has explored:

- Fixed-time and actuated signal control
- Centralized AI-based optimization systems
- Single-agent reinforcement learning models

However, key gaps remain:

- Limited real-world applicability of centralized approaches
- Poor scalability in large, multi-intersection networks
- Insufficient exploration of decentralized coordination
- Lack of practical simulation-to-real-world transition frameworks

This research aims to address these gaps through a **decentralized MARL framework**, where coordination emerges through interaction rather than explicit control.

5. Proposed Approach

Core Idea

Each traffic intersection is modeled as an independent **learning agent**.

- Agents observe local traffic conditions
- Agents take actions (signal phase decisions)
- Agents receive rewards based on traffic efficiency
- Collective behavior emerges from decentralized learning

6. Methodology

6.1 Simulation Environment

- Grid-based traffic network (e.g., 4×4 intersections)
- Configurable traffic density and flow patterns
- Real-time vehicle movement modeling

6.2 Baseline System

- Fixed-time signal control
- Predefined signal cycles
- Used as a benchmark for comparison

6.3 MARL Framework (In Progress)

State Space

- Queue length per lane
- Waiting time
- Traffic density

Action Space

- Signal phase selection
- Phase duration adjustment

Reward Function

- Minimize waiting time
- Minimize queue length
- Maximize throughput

Learning Strategy

- Decentralized training per agent
- Exploration vs exploitation balance
- Potential use of:
    - Q-Learning
    - Deep Q Networks (DQN)
    - Policy Gradient methods

7. Evaluation Metrics

The system will be evaluated using:

- Average waiting time
- Average queue length
- Throughput (vehicles cleared per unit time)
- Congestion index

Comparative analysis will be performed between:

- Fixed-time baseline system
- MARL-based adaptive system

8. Expected Contributions

This research is expected to contribute:

- A scalable MARL framework for traffic signal optimization
- A comparative evaluation of decentralized vs traditional systems
- Insights into emergent coordination in multi-agent environments
- A simulation platform for future research in intelligent transportation systems

9. Research Roadmap

Phase 1 (Completed)

- Simulation environment development
- Baseline system implementation

Phase 2 (Ongoing)

- MARL model design
- Agent architecture definition
- Initial training experiments

Phase 3

- Model optimization
- Hyperparameter tuning
- Multi-scenario evaluation

Phase 4

- Scalability testing
- Integration with real-world datasets

Phase 5

- Research publication
- Potential real-world deployment exploration

10. Current Status

- Simulation environment established
- Baseline traffic system implemented
- Evaluation framework under active development
- MARL system under active development

11. Potential Impact

Urban Mobility

- Reduced congestion and travel time

Energy Efficiency

- Lower fuel consumption and idle time

Environmental Benefits

- Reduced emissions and improved air quality

Smart Cities

- Scalable and adaptive infrastructure
- Integration with IoT and future mobility systems

12. Collaboration Interest

This research is being developed with the intention of evolving into a formal academic contribution.

The author is interested in collaborating with academic institutions and research groups to:

- Expand the work with academic rigor
- Validate the approach using real-world data
- Contribute to publishable research in intelligent transportation systems

Flexible collaboration models, including part-time or industry-aligned research pathways, are welcomed.

13. Conclusion

This research represents an effort to move beyond static and reactive traffic systems toward **intelligent, adaptive, and decentralized urban infrastructure**.

By leveraging Multi-Agent Reinforcement Learning, the study aims to demonstrate how cities can transition toward **self-optimizing traffic ecosystems**, improving efficiency, sustainability, and overall quality of urban life.